K J SOMAIYA INSTITUTE OF TECHNOLOGY

Department of Information Technology

Academic Year: 2026-27
Class: S.Y. B.Tech.
Semester: III

Subject: DSA LAB
Course Code: ITL302

Name: Rudra Desai
Division: IT-A
Batch: SA2
Roll Number: 25

Experiment No:1

Aim: Implementation of Stack using Array for real-world application.

Program:

```
#include <stdio.h>
int stk[10],i,top=-1,ele,n,choice;

void pop(){
   if(top<0){
      printf("\nStack Underflow(No active alerts)\n");
   }
   else{
      ele=stk[top];
      top--;
      printf("\nPopped Alert Element: %d\n",ele);
   }
}

void push(){
   if(top>=n-1){
      printf("\nStack Overflow\n");
   }
   else{
      printf("\nEnter the temperature to be pushed: ");
      scanf("%d", &ele);
      if (ele >= 45) {
         printf(">>> ALERT: Red Alert! Extreme Heatwave <<<\n");
      }
      else if (ele >= 43) {
         printf(">>> ALERT: Orange Alert! Severe Heatwave <<<\n");
      }
      else if (ele >= 40) {
         printf(">>> ALERT: Yellow Alert! Heatwave Warning <<<\n");
      }
      else {
         printf("\nTemperature is normal (< 40 deg). Clearing all active alerts...\n");
         while (top >= 0) {
            pop();
         }
```

```
            return;
        }
        top++;
        stk[top]= ele;
    }
}

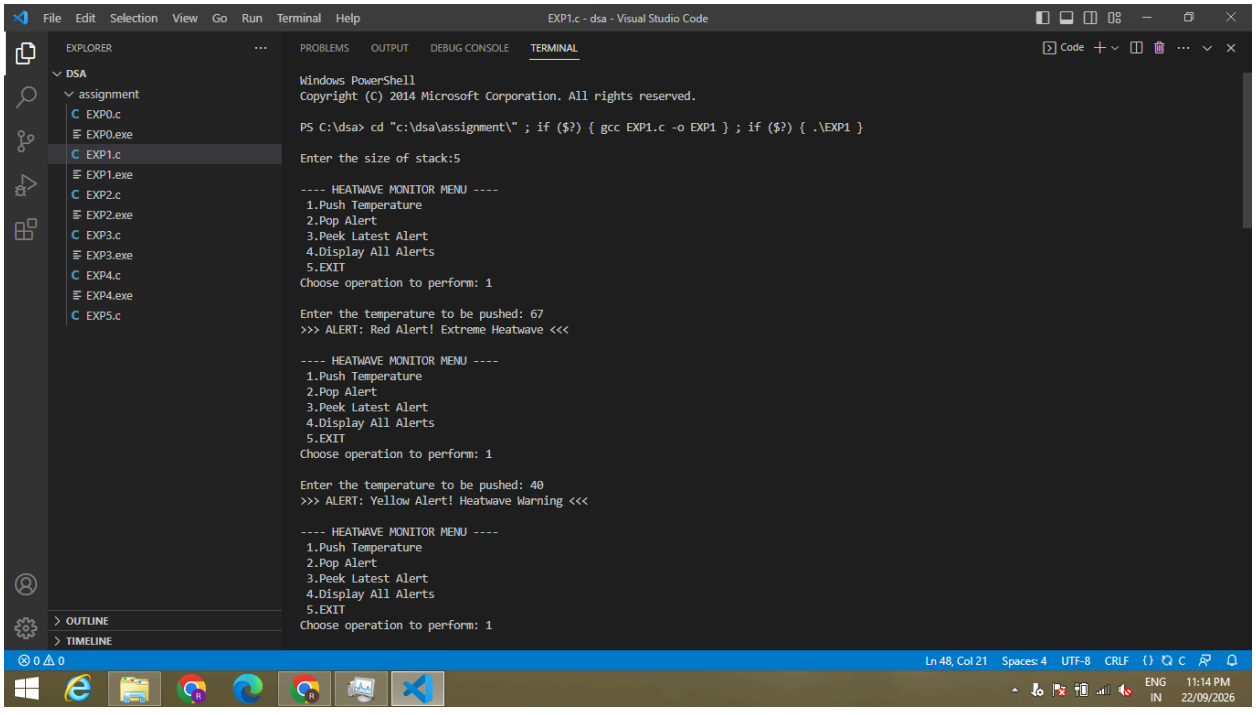
void peek(){
    if(top<0){
        printf("\nStack is Empty (No active alerts)\n");
    }
    else{
        ele= stk[top];
        printf("\nTop Alert Element: %d\n",ele);
    }
}
void display(){
    if(top<0){
        printf("\nStack is Empty (No active alerts)\n");
    }
    else{
        printf("\nACTIVE ALERTS STACK:");
        for(i=top;i>=0;i--){
            printf("\n %d", stk[i]);
        }
    }
}

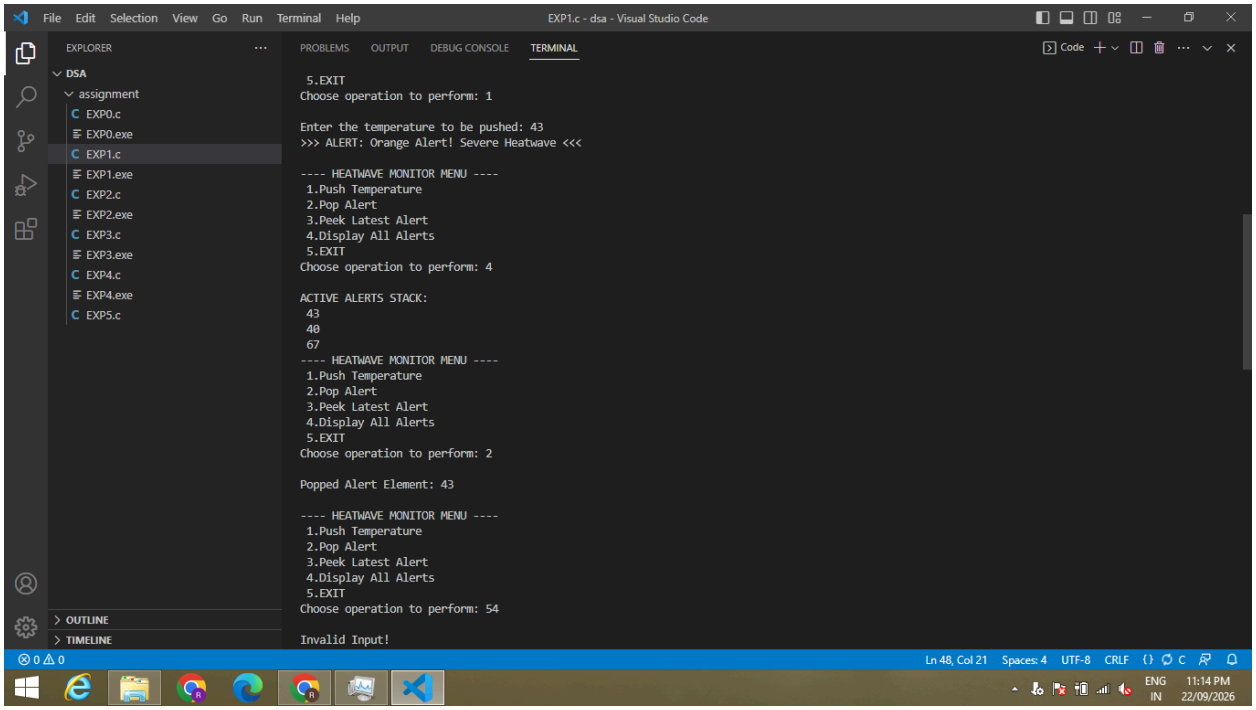
void main(){
    printf("\nEnter the size of stack:");
    scanf("%d", &n);
    do
    {
        printf("\n---- HEATWAVE MONITOR MENU ----\n 1.Push Temperature \n 2.Pop
Alert \n 3.Peek Latest Alert \n 4.Display All Alerts \n 5.EXIT");
        printf("\nChoose operation to perform: ");
        scanf("%d", &choice);
        switch (choice)
        {
        case 1 :
```

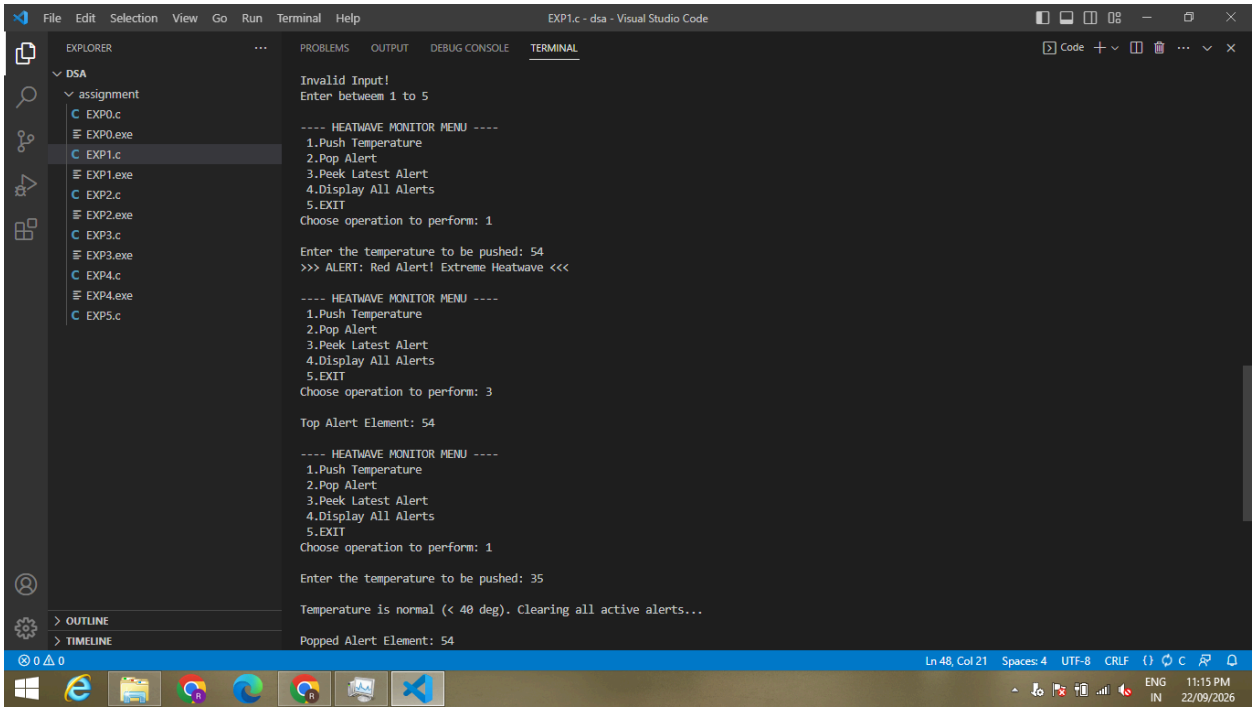
```
            push();
            break;
        case 2:
            pop();
            break;
        case 3:
            peek();
            break;
        case 4:
            display();
            break;
        case 5:
            printf("\nExited Successfully!!!\n");
            break;
        default:
            printf("\nInvalid Input!\nEnter betweem 1 to 5\n");
            break;
        }
    } while (choice !=5);

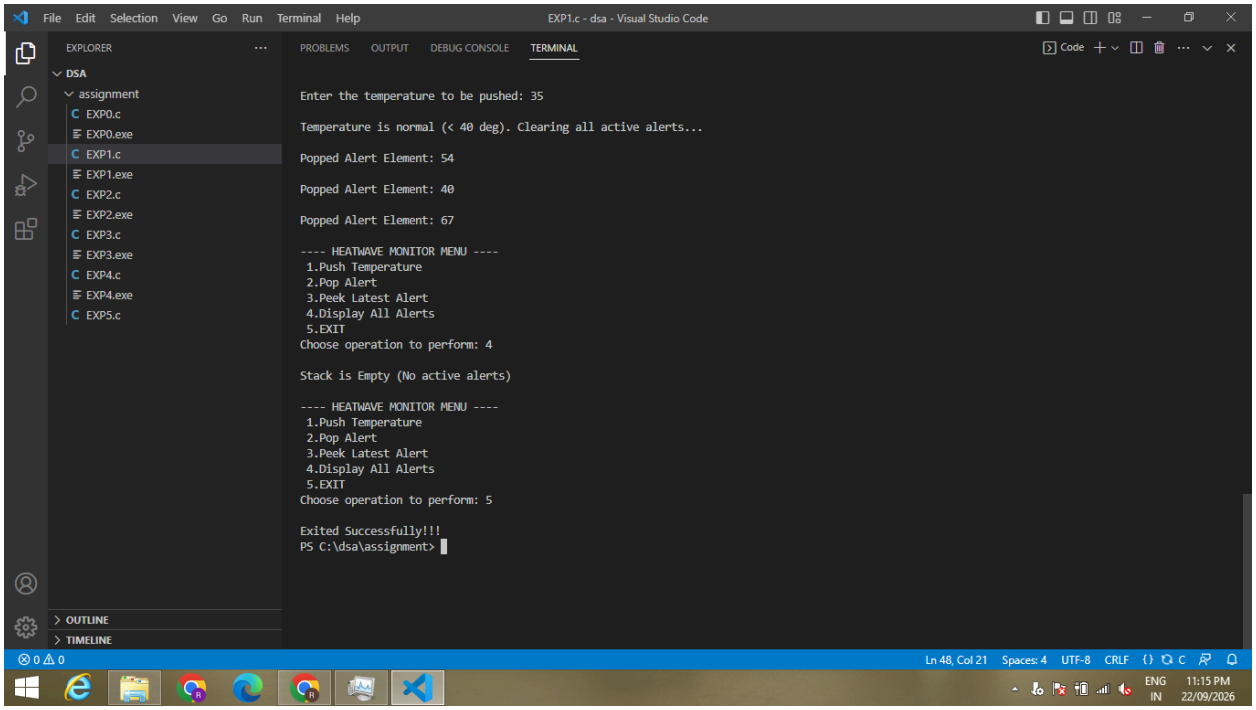
}
```

Outcome:




File Edit Selection View Go Run Terminal Help
EXP1.c - dsa - Visual Studio Code
EXPLORER
DSA
assignment
EXP0.c
EXP0.exe
EXP1.c
EXP1.exe
EXP2.c
EXP2.exe
EXP3.c
EXP3.exe
EXP4.c
EXP4.exe
EXP5.c
OUTLINE
TIMELINE
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
Code
5.EXIT
Choose operation to perform: 1
Enter the temperature to be pushed: 43
>>> ALERT: Orange Alert! Severe Heatwave <<<
---- HEATWAVE MONITOR MENU ----
1.Push Temperature
2.Pop Alert
3.Peek Latest Alert
4.Display All Alerts
5.EXIT
Choose operation to perform: 4
ACTIVE ALERTS STACK:
43
40
67
---- HEATWAVE MONITOR MENU ----
1.Push Temperature
2.Pop Alert
3.Peek Latest Alert
4.Display All Alerts
5.EXIT
Choose operation to perform: 2
Popped Alert Element: 43
---- HEATWAVE MONITOR MENU ----
1.Push Temperature
2.Pop Alert
3.Peek Latest Alert
4.Display All Alerts
5.EXIT
Choose operation to perform: 54
Invalid Input!
Ln 48, Col 21 Spaces: 4 UTF-8 CRLF C
ENG IN 11:14 PM 22/09/2026


File Edit Selection View Go Run Terminal Help
EXP1.c - dsa - Visual Studio Code
EXPLORER
DSA
assignment
EXP0.c
EXP0.exe
EXP1.c
EXP1.exe
EXP2.c
EXP2.exe
EXP3.c
EXP3.exe
EXP4.c
EXP4.exe
EXP5.c
OUTLINE
TIMELINE
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
Code
Invalid Input!
Enter betweem 1 to 5
---- HEATWAVE MONITOR MENU ----
1.Push Temperature
2.Pop Alert
3.Peek Latest Alert
4.Display All Alerts
5.EXIT
Choose operation to perform: 1
Enter the temperature to be pushed: 54
>>> ALERT: Red Alert! Extreme Heatwave <<<
---- HEATWAVE MONITOR MENU ----
1.Push Temperature
2.Pop Alert
3.Peek Latest Alert
4.Display All Alerts
5.EXIT
Choose operation to perform: 3
Top Alert Element: 54
---- HEATWAVE MONITOR MENU ----
1.Push Temperature
2.Pop Alert
3.Peek Latest Alert
4.Display All Alerts
5.EXIT
Choose operation to perform: 1
Enter the temperature to be pushed: 35
Temperature is normal (< 40 deg). Clearing all active alerts...
Popped Alert Element: 54
Ln 48, Col 21 Spaces: 4 UTF-8 CRLF C
ENG IN 11:15 PM 22/09/2026


File Edit Selection View Go Run Terminal Help
EXP1.c - dsa - Visual Studio Code
EXPLORER
DSA
assignment
EXP0.c
EXP0.exe
EXP1.c
EXP1.exe
EXP2.c
EXP2.exe
EXP3.c
EXP3.exe
EXP4.c
EXP4.exe
EXP5.c
OUTLINE
TIMELINE
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
Code
Enter the temperature to be pushed: 35
Temperature is normal (< 40 deg). Clearing all active alerts...
Popped Alert Element: 54
Popped Alert Element: 40
Popped Alert Element: 67
---- HEATWAVE MONITOR MENU ----
1.Push Temperature
2.Pop Alert
3.Peek Latest Alert
4.Display All Alerts
5.EXIT
Choose operation to perform: 4
Stack is Empty (No active alerts)
---- HEATWAVE MONITOR MENU ----
1.Push Temperature
2.Pop Alert
3.Peek Latest Alert
4.Display All Alerts
5.EXIT
Choose operation to perform: 5
Exited Successfully!!!
PS C:\dsa\assignment>
Ln 48, Col 21 Spaces: 4 UTF-8 CRLF C
ENG IN
11:15 PM 22/09/2026